

Delayed Discounting Experiment

PSY310

Lab in Psychology Lab Report

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GitHub Link: <https://github.com/saiman-ui/Lab-Psychology/tree/Delayed-Discounting>

Introduction

Human choice-making tends to be deciding between smaller rewards now and bigger rewards later — a process referred to as delay discounting. Delay discounting tasks quantify the way people discount later outcomes compared to sooner ones.

Theoretically, this is a function of differences in self-control and impulsivity. Hyperbolic discounting models suggest that the subjective value of delayed rewards declines non-linearly with increasing time delay (Mazur, 1987). Highly impulsive individuals are more likely to opt for immediate rewards and thus exhibit higher discounting rates (k-values).

Earlier research (e.g., Kirby et al., 1999; Bickel et al., 2011) has established large interindividual differences in discounting rates that relate to such behaviors as drug use, procrastination, and financial choices. This experiment sought to quantify these individual differences among participants with a minimal two-alternative forced-choice task created using PsychoPy.

Method

Participant

The task was performed by one participant, a 20-year-old female undergraduate student at Ahmedabad University. Prior to the start of the study, she signed informed consent. She had to determine whether a sinusoidal grating stimulus was tilted (signal present) or vertical (signal absent).

Materials and Procedure

Four participants (counterbalanced using the same experimental apparatus) performed a computerized delay discounting task. Each participant made 100 choices between a small reward paid immediately ("today") and a larger delayed reward ("later").

The experiment was prepared using PsychoPy Builder. Each trial showed a fixation cross ("+") for one second, followed by a choice screen like:

"₹11 today or ₹30 after 7 days"

Participants answered by pressing:

'T' for "today" (immediate reward)

'L' for "later" (delayed reward)

One hundred combinations of reward and delay values in the form of a CSV file were used to randomize trials. Responses of each participant were automatically saved as a .csv file.

Each participant performed 100 randomized trials. PsychoPy data was exported to Microsoft Excel for analysis. In each trial, the discounting rate (k) was obtained by applying the hyperbolic discounting equation:

$$k = \frac{V - DA}{V}$$

The geometric mean of k-values for trials where participants chose the *delayed* option was used as the participant's overall discounting rate.

Results

Participant	Mean (k-Value)	Interpretation
P1	0.371	Moderately Impulsive
P2	0.283	Somewhat Impulsive
P3	0.537	Highly Impulsive
P4	0.097	Self-Controlled

Group Geometric Mean:

$$K_{\text{group}} = (0.371 * 0.283 * 0.537 * 0.097)^{1/4} = 0.246$$

Thus, the overall group discounting rate was $k = 0.25$ (approx).

Discussion

Results show within-subject variation in delay discounting. Participant 3 registered the highest k-value (0.537), which suggests more preference for immediate reward — greater impulsivity. Participant 4, with a considerably lower k-value (0.097), was inclined to wait more for delayed rewards, which suggests greater self-control.

The group's geometric mean ($k=0.25$) indicates a moderate rate of impulsivity across the board, consistent with past studies demonstrating average human preference for smaller sooner rewards when uncertain.

Reference

- Bickel, W. K., et al. (2011). The behavioral economics of drug dependence: Toward the consilience of economics and neuroscience. *Current Directions in Psychological Science*, 20(5), 265–269.
- Kirby, K. N., Petry, N. M., & Bickel, W. K. (1999). Heroin addicts have higher discount rates for delayed rewards than non-drug-using controls. *Journal of Experimental Psychology: General*, 128(1), 78–87.
- Mazur, J. E. (1987). An adjusting procedure for studying delayed reinforcement. In *Quantitative analyses of behavior* (Vol. 5, pp. 55–73).